

PRIME EQUIPMENT GROUP

JBT EMS Efficiency Management System

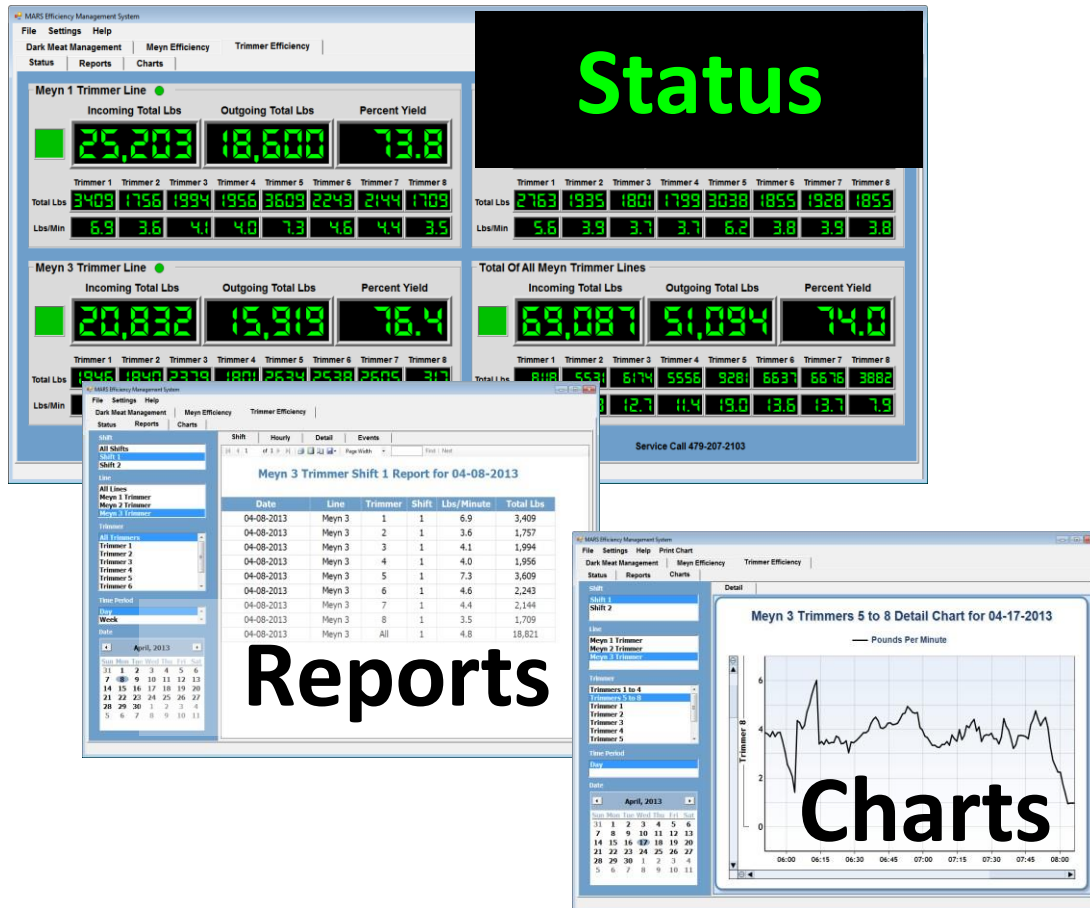


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MARS EMS Introduction

The MARS Efficiency Management System (**EMS**) helps monitor and manage efficiency in a poultry processing plant. The system offers live status screens, popup alarms, on-demand reports, and scheduled reports. The system highlights high-profile issues such as:

- Efficiency targets dropping below a certain level.
- Unexpected line downtimes.
- Communication issues with MARS controllers.

Reports and Charts

- **Hourly and Shift Production Reports:** Summarize production data.
- **Detailed Reports:** Track production minute by minute.
- **Events Report:** Logs key events like line downtimes, starts, and no product on the line.
- **Detailed charts:** Shows minute-by-minute production during shifts, helping identify events like line speed changes, stoppages, or significant drops in product rate.

Modules

The MARS EMS modules interface with the different types of JBT controllers in a poultry processing facility. Each module is discussed in more detail.

- Counters
- Scales
- Paw Tracking and Rejection
- Scalders
- Trimmer Efficiency.

Automatic Notifications

Automatic emails and text messages can be sent to recipients during the production day. These include:

- End of shift, hourly, lunch, and weekly reports.
- Special events related to specific modules.

For example:

- **Counters Module:** Alerts for line downtimes or no product on a line.
- **Scalders Module:** Alerts for out-of-range temperatures or critical errors in the MARS Scalders controller.

Required Software

- **Server Version:**
 - Must be installed on the computer connected to MARS Controllers.
 - Communicates with controllers and stores data in a MS SQL Server database.
 - Used for scheduling and settings changes.
 - Supports testing communications with controllers.
- **Client Version:**
 - Can be installed on other computers in the plant.
 - Provides access to live status screens, reports, and charts.
 - Requires the Server and Client computers to be on the same network.
 - VNC can be used to connect to the Server from other locations.

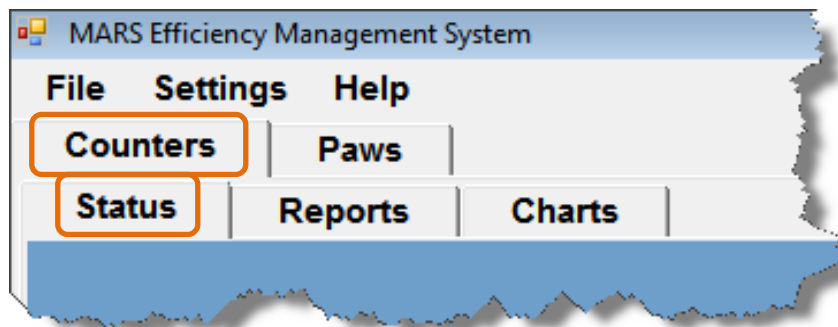
Main Window

The MARS Efficiency Management System is a Microsoft Windows application with a menu bar featuring **File**, **Settings**, and **Help** options.

- **Primary Tabs:** Each module has a primary tab. For example, **Counters** and **Paws** indicate installed modules.
- **Subtabs:** Each primary tab has subtabs specific to it.
- **Report Tabs:** Individual report selections located directly above the report window.

Example:

- **Counters Tab:** Subtabs include **Status**, **Reports**, and **Charts**.
- **Paws Tab:** Subtabs include **Status** and **Reports** (no **Charts** subtab)



Counters Tab Selected Above



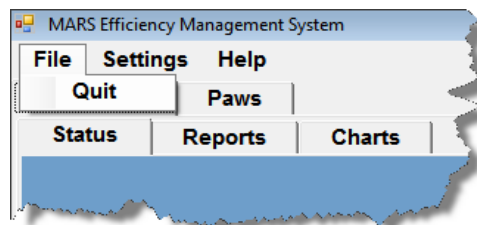
Paws Tab Selected Above

Menu Bar

The MARS Efficiency Management System has a standard menu bar with **File**, **Settings** and **Help** options. Each menu option will be discussed in detail on the following pages.

File Menu

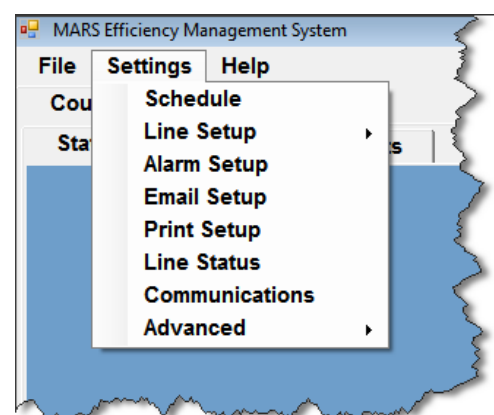
- Quit / Exit the program



Settings Menu

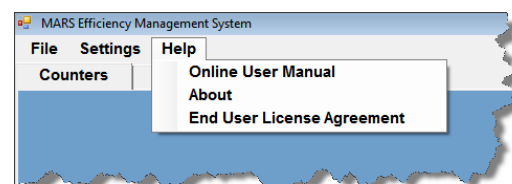
The primary menu that is used for initial and advanced setup configuration.

- **Schedule**
 - Manage production periods
- **Line Setup**
 - Counters
 - Scalders
- **Alarm Setup**
 - Configure popup alarms
- **Email Setup**
 - Manage recipient addresses
 - Assign reports
 - Sntp server settings
- **Print Setup**
 - Set print time options
- **Line Status**
 - Turn lines on/off
 - Set line end times
- **Communications**
 - Configuration
 - Test communications
 - Reset controllers
- **Advanced**
 - Counter Controller Setup



Help Menu

- Online User Manual
- About
- End User Agreement
 - Required Acceptance: Please review and accept the agreement, or **File - Quit** to exit the program



Settings - Schedule

Select the **Settings** menu option from the main window, then the **Schedule** menu option.

To create or edit a production line schedule:

1. Choose the line from the **Select Line to Change** dropdown.
2. Pick the number of production periods from the **Select the Number of Production Periods** dropdown.

A production period is active production time, excluding breaks, lunch, or scheduled downtime. It usually starts at the beginning of a shift or after a break and ends at the start of a break or the end of a shift.

The screenshot shows the 'Scheduler' window with the following details:

- Production Periods** section:
 - Select Line to Change:** Kill Line
 - Select the Number of Production Periods:** 6
- Production Periods Table:**

Production Period 1	Production Period 2	Production Period 3	Production Period 4	Production Period 5	Production Period 6
Start: 07/18/13 06:00	Start: 07/18/13 08:45	Start: 07/18/13 11:45	Start: 07/18/13 15:00	Start: 07/18/13 17:45	Start: 07/18/13 20:45
Stop: 07/18/13 08:30	Stop: 07/18/13 11:15	Stop: 07/18/13 14:45	Stop: 07/18/13 17:30	Stop: 07/18/13 20:15	Stop: 07/18/13 23:45
☒ New Shift	☐ New Shift	☐ New Shift	☒ New Shift	☐ New Shift	☐ New Shift
- Global Settings:**
 - ☒ Make All Schedule Changes Permanent
 - ☒ Copy Line 1 Schedule To All Lines
- Buttons:** OK, Cancel, Apply

The beginning of a **Shift** is indicated by a check box named **New Shift** for a particular production period. If it is checked, then it marks the beginning of a new shift and if unchecked the production period is not the beginning of a shift.

By default, all changes made to schedules are saved permanently. If changes to the schedules are meant to be temporary, then uncheck the **Make All Schedule Changes Permanent** checkbox.

If you are updating the schedule for line 1 and you would like the changes to be applied to all production lines, then check the **Copy Line 1 Schedule To All Lines** checkbox. If you are editing the schedule for any line other than line 1 and this checkbox is checked, then a warning message will be displayed indicating that you must be updating line 1 for this option to be relevant.

Settings – Schedule - Edit Production Period

To change a production period, start and stop time, click on the production period to update from the Scheduler screen and the **Edit Production Period** screen will be displayed as shown below. You can then update the **Start Time** and **Stop Time** using the **Hour** and **Minute** dropdown lists.

Section	Field	Value
Start Time	Month	7
	Day	23
	Year	2013
	Hour	15
	Minute	00
	Buttons	Edit Dates, OK, Cancel
Stop Time	Month	7
	Day	23
	Year	2013
	Hour	17
	Minute	30
	Buttons	Edit Dates, OK, Cancel

If you need to change the Date for the Start Time or the Stop Time because the production period overlaps a day boundary, then click on the **Edit Dates** button. This will enable the Month, Day and Year dropdown lists so you can update them appropriately.

After editing the production period times and optionally the dates, click on the **OK** button to save your changes.

Be careful to avoid overlapping production periods when you are updating the schedules.

When you are satisfied with your new schedule, save it by clicking the **Apply** button or the **OK** button. If you want to discard all of your changes, click the **Cancel** button. Clicking the **Apply** button instead of the **OK** button allows you to save the schedule while keeping the Scheduler open to continue editing. Clicking the OK button will close the **Scheduler** screen in addition to saving your changes.

Settings – Line Setup - Counters

To access the Counter Line Setup screen, choose Line Setup and then Counters from the Settings menu.

In the Counter Line Setup screen, you can:

- Set the projected Product Rate Per Minute.
- Configure thresholds for Line Speed, Product Rate, Line Efficiency, and Product Efficiency.

These thresholds determine the color (Green, Yellow, Red) shown on the Counters Status tab, indicating how well production meets targets. Some setups also have Product Rate lights on the production floor to alert staff when targets aren't met.

Select the line to configure from the dropdown list below **Select Line**. Then, select the desired Product Target Rate Per Minute from the corresponding dropdown list.

Set the Yellow and Red Threshold values for Line Speed and Product Rate below the Product Target Rate Per Minute.

- **Red Thresholds:** Lower than Yellow, indicating poorer performance.
- **Yellow Indicators:** Appear when values fall below Yellow Thresholds.
- **Red Indicators:** Appear when values fall below Red Thresholds.

Configure Line Efficiency and Product Efficiency thresholds similarly but use percentages instead of Product Rate Per Minute. Again, set Red Thresholds lower than Yellow.

When finished, click on either the **Apply** button or the **OK** button to save your changes. To discard your changes, click the **Cancel** button.

Settings – Line Setup - Scalders

To access the Scalders Line Setup screen, select Line Setup and then Scalders from the Settings menu.

Temperature thresholds are set on this screen. These thresholds determine the color (Green, Yellow, Red) shown on the Scalders Status tab, indicating how close the actual temperature is to the desired set temperature.

Select the line to configure from the dropdown list below **Select Line**.

Then configure the number of degrees the actual temperature is from the desired set temperature values that will trigger Yellow and Red for each system. **The Red threshold values should be set to higher values than the corresponding yellow threshold value.**

To apply scalders threshold changes from line 1 to all lines, check the **Copy Line 1 Settings To All Lines** checkbox.

If you're updating thresholds for a line other than line 1 and this box is checked, a warning will appear, indicating that you must be updating line 1 for this option to work.

When finished, click on either the **Apply** button or the **OK** button to save your changes. To discard your changes, click the **Cancel** button.

Settings - Alarm Setup

The **Alarm Setup** screen is displayed by selecting **Alarm Setup** from the **Settings** menu on the main screen. The **Alarm Setup** screen as shown on the top of the next page is where you configure what events will trigger popup alarms on the screen and whether the popup alarms will include flashing and audio beeps to better alert personnel.

In the **Alarm Events** window, you can turn alarms on or off:

- **Check/Uncheck:** Click to enable or disable events. A checkmark means the alarm is on.

In **Alarm Settings:**

- Choose if popups should flash or beep using checkboxes.
- Set the time in seconds between flashes and beeps. Adjust the values if needed.

Alarms can be turned on or off by clicking either **Alarms On** or **Alarms Off** at the bottom left of the screen.

Click the **OK** button when finished to save your changes or click the **Cancel** button to discard your changes.

Sample popup alarm indicating that the Product Rate for the Kill Line has dropped into the red threshold. This alert can be used to make personnel aware of the drop in production so they can investigate the cause.

Sample a popup alarm alerting that the Paw Line controller is not communicating with the MARS Efficiency Management System. This alert should trigger maintenance personnel to investigate and fix the problem.

Settings - Email Setup

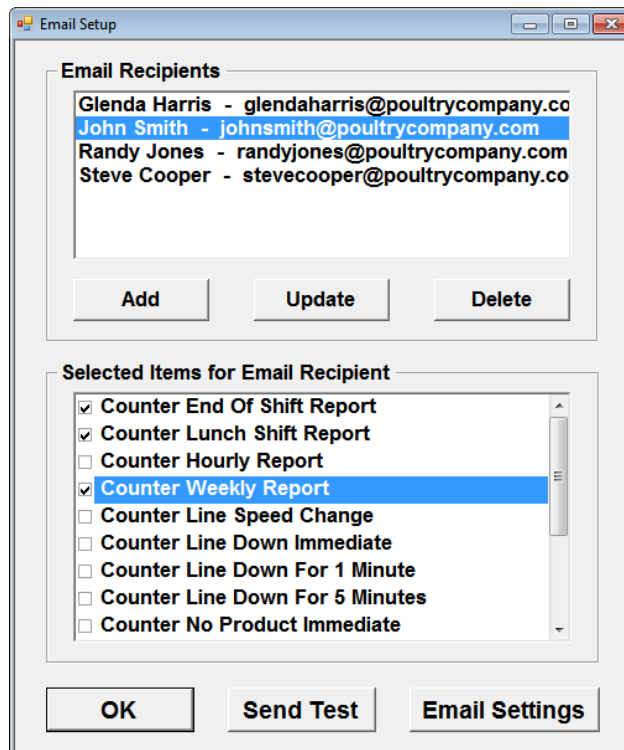
Access **Email Setup** by selecting **Settings** menu on the main screen, then **Email Setup** menu option. Use this screen to manage email recipients of EMS reports.

Click on each name in the **Email Recipients** window to see which items are checked in the **Selected Items for Email Recipient** window.

The items that have a checkmark to the left indicate which emails will be automatically sent to the recipient.

Click to check or uncheck the email items for the selected recipient as desired.

Scroll down to see additional email items that can be configured.



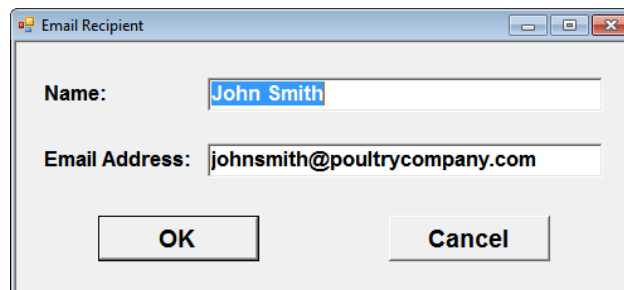
Settings - Email Recipient

Click the **Add/Update** button on the previous screen to manage an email recipient's contact information. The **Email Recipient** window will be displayed as shown below.

Enter/Update the recipient's name and email address.

To send events as text messages to cell phones, type the phone number with the correct suffix for your mobile service provider as the email address.

The suffix for AT&T is @txt.att.net,
Verizon @vtext.com,
Sprint @messaging.sprintpcs.com,
T-Mobile @tmomail.net.



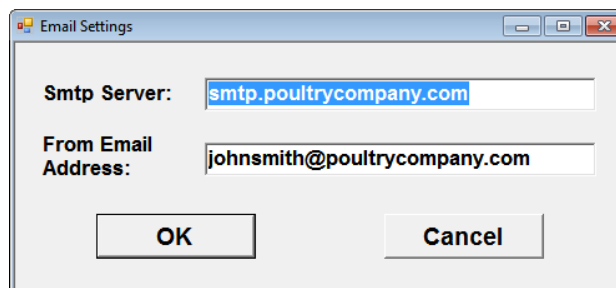
Do not include dashes in the phone number portion.

Settings – Email Recipient - Email Settings

Click the **Email Settings** button at the bottom right of the **Email Recipients** screen to display the screen shown below.

Change the name of the outgoing **SmtP Server** to match the name of the server your company uses for outgoing email.

Type the email address that you would like to have used for the **From Email Address** when emails are automatically sent by the MARS EMS System.



The 'Email Settings' dialog box contains two text input fields. The first field, labeled 'SmtP Server:', contains the text 'smtp.poultrycompany.com'. The second field, labeled 'From Email Address:', contains the text 'johnsmith@poultrycompany.com'. At the bottom of the dialog are two buttons: 'OK' and 'Cancel'.

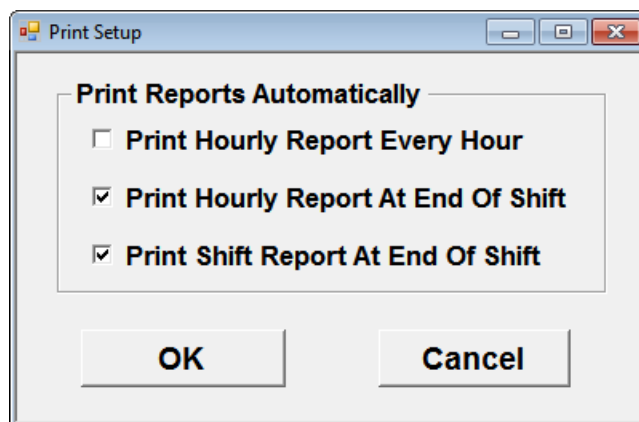
Settings - Print Setup

The **Print Setup** screen shown below is displayed by selecting **Print Setup** from the **Settings** menu on the main screen. Use the **Print Setup** screen to configure which reports will be printed automatically during the day.

Click to check or uncheck the reports that should be printed automatically.

A checkmark to the left of the item indicates that the item is turned on for automatic printing.

Click the **OK** button when finished to save your changes or click the **Cancel** button to discard your changes.



The 'Print Setup' dialog box features a section titled 'Print Reports Automatically'. Below this title are three items, each with a checkbox: 'Print Hourly Report Every Hour' (unchecked), 'Print Hourly Report At End Of Shift' (checked), and 'Print Shift Report At End Of Shift' (checked). At the bottom of the dialog are two buttons: 'OK' and 'Cancel'.

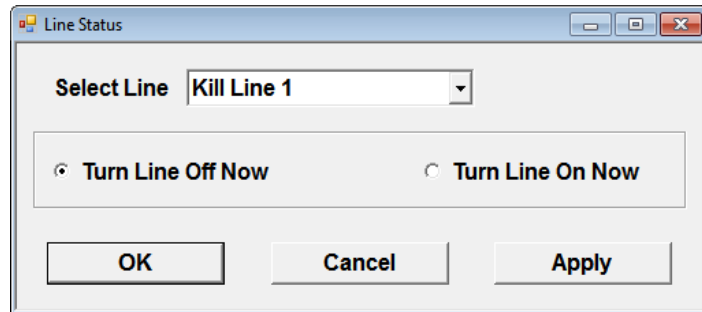
Settings - Line Status

The **Line Status** screen shown below is displayed by selecting **Line Status** from the **Settings** menu on the main screen. The **Line Status** screen allows you to turn a line on or off during the day. Turning a line off when it is not being used will ensure that the shift report will be accurate.

Select the line to turn off or on from the dropdown list to the right of **Select Line**.

Click **Turn Line Off Now** or **Turn Line On Now** as appropriate.

Turned Off Lines are turned on again at the beginning of the next shift.



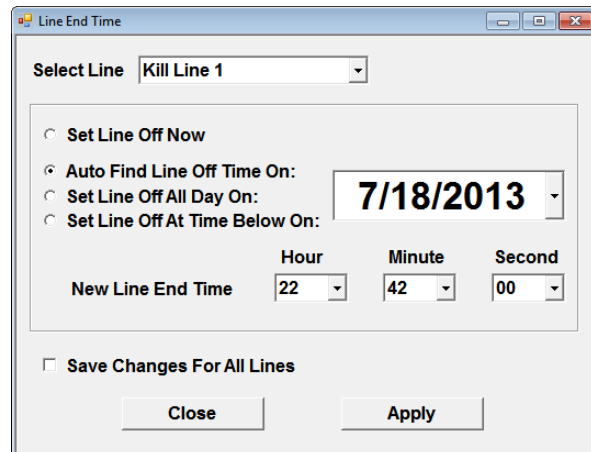
Line End Time

The **Line End Time** screen shown below is accessed by using the keyboard combination **Ctrl + T**. The **Line End Time** screen allows you to turn a line off now for the current day or allows you to go back and update the line end time for a previous day.

This screen is typically used when the last shift of the day doesn't always end at the same time.

The end time for the last shift of the day should be set in the Scheduler to the latest time that the line could end for the day. This ensures that the reports will keep recording data until the latest time a shift could end.

The MARS software automatically tries to figure out the actual shift end time by examining the detail data for a line, but this screen can be used to override the actual end time if it is not correct. This ensures accuracy on the final totals and efficiencies on reports and charts.



Select the appropriate line in the Select Line dropdown list and click on the desired change and set the **New Line End Time** if applicable.

Click the checkbox named **Save Changes For All Lines** if you would like to update all of the lines and then click the **Apply** button to save your changes. When finished, click the **Close** button. *Lines that are turned off now for the current day will be automatically turned on again at the beginning of the next shift.*

Settings - Communications

The **Communications** screen as shown below allows the user to change and test the communication settings. The communication settings control how the Windows computer running the MARS Efficiency Management System exchanges data with the controllers out in the production area. To display the screen below, select the **Communications** option from the **Settings** menu. Type in the appropriate password if prompted and click the **OK** button to display the Communications screen shown below. The password is typically **admin1**.

Communication Settings

The Communications screen provides for setting the Port, Maximum Speed or Baud Rate, Number of Data Bits, Odd or Even Parity, and the Number of Stop Bits. The program default settings are typically correct and normally do not need to be changed except for the Port value which is specific to each computer.

Chose Ok/Cancel/Apply when finished.

The **Apply** button keeps the screen open for additional testing and **OK** saves and closes the window.

Communication Test

Use the Communications screen to test controller communications on the production floor. Each controller has been assigned a unique Modbus communication address.

Single Test:

1. Select a Modbus Address next to the 'Click to Test Communications' button.
2. Click the button to test.
3. View results in the upper right window. (i.e. Product Count, Product Rates, or Errors)

Continuous Test:

1. Select the Modbus Addresses to test.
2. Click Continuous Scan Mode to start test.
4. View results in the upper right window, . (i.e. Product Count, Product Rates, or Errors)
3. Click Scan Disable to stop the test.
4. Use the Clear button to erase results before a new test if needed.

Manual Reset Controllers

Use the **Communications** screen to manually reset totals for a controller. The system automatically resets Totals at the start of each shift.

- **Manual Reset Single Controller:** Select the Modbus Address next to the 'Click to Test Communication' button and click **Reset Selected**.
- **Manual Reset All Controllers:** Click **Reset All**.

Settings – Advanced - Counter Controller Setup

Select Settings menu on the main screen, then Advanced menu option, and then **Counter Controller Setup**. This screen is only applicable to installations that include the Counter Efficiency module and is used to display and change the **Sensor Minimum Block Times** on a MARS Controller.

The **Shaft Sensors** and **Product Sensors Primary** values are the minimum block times in milliseconds that the Primary sensors must be blocked to be considered valid. If the MARS Controller includes a second set of **Redundant Sensors** for comparison counts, then these values will be displayed as well.

Select the line to display and change values for from the dropdown list to the right of **Select Line**. The values will then be read and displayed from the appropriate MARS Controller.

The **Status** will be updated to display either “Counter Settings Loaded Successfully” or “Error Loading Counter Settings” if an error occurred.

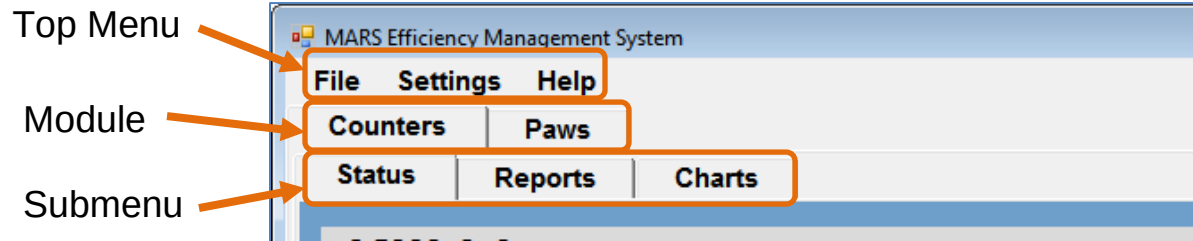
The screenshot shows a window titled "Counter Controller Setup" with a subtitle "Sensor Minimum Block Times". It features a dropdown menu for "Select Line" currently set to "Kill Line". Below this, a status message reads "Status: Counter Settings Loaded Successfully". The main configuration area has two columns: "Primary" and "Redundant". Under "Primary", "Shaft Sensors" is set to 20 and "Product Sensors" is set to 100. The "Redundant" column has empty input fields for both sensor types. At the bottom, there is a checkbox labeled "Save Changes For All Lines" which is unchecked, and three buttons: "OK", "Cancel", and "Apply".

Change the minimum block time values if desired and click the checkbox named **Save Changes For All Lines** if you would like to update all of the lines with the new values.

Chose Ok/Cancel/Apply when finished. The **Apply** button keeps the screen open for additional testing and **OK** saves changes and closes the window.

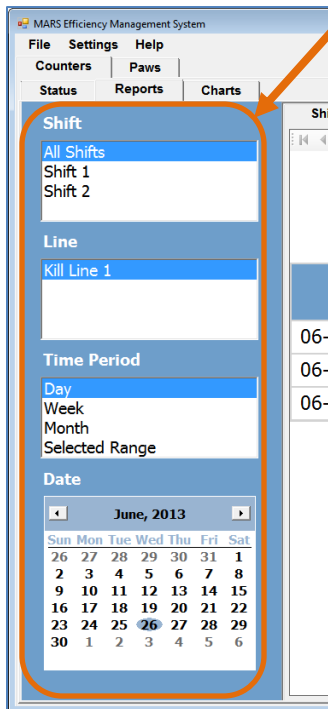
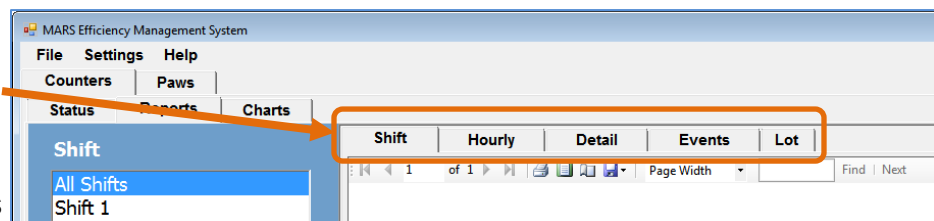
MARS EMS Navigation

The MARS EMS Status, Report and Charts navigations work the same across all modules.



Report Menu

Report options may vary between modules



Report Filters

The blue panel on the left side of the report window contains the filters that can be applied to the report. Select the desired filter in each list. Available time periods are Day, Week, Month, or Selected Range.

How to **Select a Range** of dates:

1. Click **Selected Range** under Time Period.
2. Click the start date on the calendar.
3. Hold the **Shift** key and click the end date.

The selected dates will be highlighted, and the report will display data for this range.

Navigating/Printing/Exporting

Standard windows icons and scrollbars are available for page navigation. Printing and exporting to various formats are available.



Counters Module

The MARS Efficiency Management System Counters Module communicates with controllers that count products on production lines or equipment like deboning machines. A single controller can handle up to four lines at once. Data is uploaded from controllers to the MARS system on a Windows computer, where the Data is stored in a Microsoft SQL Server database and accessed via live Status, Reports, and Charts tabs.

Report Generation and Navigation

The available reports and charts follow the general navigation shown at the beginning of this document. Available reports are listed on the report tab and report filters are on the left blue panel. General windows navigation and printing/exporting options are above the report.

How to enter a range of dates:

To use the **Selected Range** option:

1. Click **Selected Range** under Time Period.
2. Click the start date on the calendar.
3. Hold the **Shift** key and click the end date.

The selected dates will be highlighted, and the report will display data for this range.

Emails and Text Messages

The Counters module can send emails and text messages for the following reports/events:

- Counter End Of Shift
- Counter Lunch
- Counter Hourly
- Counter Weekly
- Counter Line Speed Change event
- Counter Line Down event
- Counter Line Down For 1 Minute event
- Counter Line Down For 5 Minutes event
- Counter No Product event
- Counter No Product For 1 Minute event
- Counter No Product For 5 Minutes event.

For detailed setup instructions, refer to the **Email Setup** and **Email Recipient** sections.

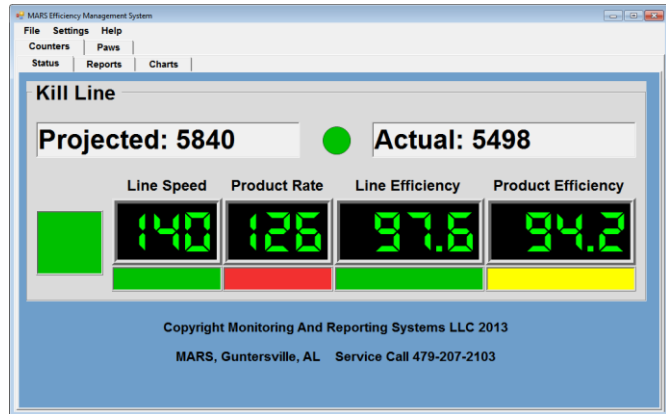
Counters - Status Tab

Select **Counters** module tab then the **Status** subtab. The **Status** screen displays the live Status of each Counter in a separate block. The screen below is for an example of one kill line counter. The **Projected and Actual Product Counts** are for the current shift.

The projected count is calculated by multiplying the total production minutes since the start of the shift by the projected product count per minute.

The projected product count per minute is configured with the **Counter Line Setup** screen as discussed previously.

Below the projected and actual counts, you can see:



- **Line Speed:** Trolleys per minute
- **Product Rate:** Products per minute
- **Line Efficiency:** For the current shift
- **Product Efficiency:** For the current shift
- **Product Rate and Line Speed:** If equal, all trolleys had products. Product rate can't exceed line speed.

Status Rectangles:

- **Green:** Value is near optimum.
- **Yellow:** Value is further from optimum.
- **Red:** Value is unacceptable.

Thresholds for these colors are set in the Counter Line Setup screen.

The counter status is shown in a colored square on the left of the counter block:

- **Green:** Line is running during active production.
- **Blue:** Line is on break, before a shift, or after a shift.
- **Red:** Line is stopped during active production.

Additional messages:

- **Line Down:** Displayed when the line is stopped.
- **No Product:** Displayed when the line is running but no product is present.

The communication status with the MARS controller is a round circle.

- **Green:** Successful communication
- **Red:** Communication failure

Counter – Reports - Shift Report

The Shift report shows the total product count, projected product count, line efficiency, product efficiency, production downtime and line downtime of each line for the selected shift, line and time period. If **All Shifts** is selected, then data for each shift will be displayed individually along with the totals for all shifts at the end.

Production Downtime is the number of minutes that have been lost or gained for any reason from what the total projected product count should be at the current time. Production minutes are lost when a line is running slower than the projected count per minute or when not all of the trolleys have product. Production minutes can be gained back when a line is running faster than the projected count per minute and every trolley has product.

Line Downtime is the number of minutes that have been lost when a line is stopped during an active production period.

The screenshot shows the 'Shift' report for 'Kill Line 1' on '06-26-2013'. The interface includes a sidebar with filters for Shift, Line, Time Period, and Date. The main table displays the following data:

Date	Line	Shift	Total Count	Projected Count	Line Efficiency	Product Efficiency	Production Downtime	Line Downtime
06-26-2013	Kill 1	1	65599	64668	99.9	101.4	-7	0
06-26-2013	Kill 1	2	60730	61931	100.0	98.1	9	0
06-26-2013	Kill 1	All	126329	126599	100.0	99.8	2	0

Counter – Reports - Hourly Report

The Hourly Report shows:

- Hourly product count
- Hourly product efficiency
- Total product count each hour during the selected shift.
- Total product efficiency for each hour during the selected shift.

The screenshot shows the 'Hourly' report for 'Kill Line 1' on '06-11-2013'. The interface includes a sidebar with filters for Shift, Line, Time Period, and Date. The main table displays the following data:

Time	Line	Shift	Hourly Count	Hourly Efficiency	Total Count	Total Efficiency
06-11 06:20:00	Kill 1	1	7450	88.7	7450	88.7
06-11 07:20:00	Kill 1	1	7799	92.8	15249	90.8
06-11 07:50:00	Kill 1	1	3725	88.6	18974	90.4
06-11 09:05:00	Kill 1	1	9801	116.8	28775	97.9
06-11 10:05:00	Kill 1	1	7824	93.2	36599	96.9
06-11 10:20:00	Kill 1	1	1833	86.7	38432	96.3
06-11 12:03:00	Kill 1	1	7773	92.6	46205	95.7
06-11 13:03:00	Kill 1	1	8330	99.2	54535	96.2
06-11 14:00:00	Kill 1	1	7476	93.6	62011	95.9

If there is a scheduled break that begins before the next hour is complete, then the report will display a partial hour ending at the start of the scheduled break. The next hourly data displayed will then be an hour after the scheduled break ends.

Counter – Reports - Detail Report

The Detail report displays the total product count, total projected count, line speed, product rate, line efficiency and product efficiency for each minute during the selected shift. During scheduled breaks an X will be displayed in the column named BK.

Time	Line	Shift	Total Count	Projected Count	BK	Line Speed	Product Rate	Line Efficiency	Product Efficiency
06-18 09:12:34	Kill 1	1	28921	30457		142	142	97.8	95.0
06-18 09:13:37	Kill 1	1	29072	30604		143	141	97.8	95.0
06-18 09:14:40	Kill 1	1	29223	30753		142	142	97.8	95.0
06-18 09:15:44	Kill 1	1	29373	30900		143	143	97.8	95.1
06-18 09:16:47	Kill 1	1	29523	31050		142	140	97.8	95.1
06-18 09:17:50	Kill 1	1	29676	31194		143	143	97.8	95.1
06-18 09:18:54	Kill 1	1	29822	31344		143	141	97.8	95.1
06-18 09:19:57	Kill 1	1	29971	31491		142	141	97.8	95.2
06-18 09:21:00	Kill 1	1	30123	31640		143	143	97.9	95.2
06-18 09:22:04	Kill 1	1	30273	31787		142	141	97.9	95.2
06-18 09:23:07	Kill 1	1	30421	31936		143	140	97.9	95.3
06-18 09:24:10	Kill 1	1	30572	32083		142	142	97.9	95.3
06-18 09:25:14	Kill 1	1	30722	32230		143	142	97.9	95.3

Counter – Reports - Events Report

The Events report has checkboxes above the report that control what events are displayed.

On the left **Line Down**, **Line Running**, **Product Started**, **Product Stopped** and **Product Resuming** events are checked.

The other events that can be displayed are

Speed Change and **Controller Resets**. The Speed Change event is for when a line changes speed and the report will show the old line speed in the **Speed Old** column and the new line speed in the **Speed New** column.

Time	Line	Shift	Description	Speed Old	Speed New
06-18 07:47:30	Kill 1	1	Line Empty / No Product	0	0
06-18 08:04:12	Kill 1	1	Product Resuming On Line	0	0
06-18 08:16:58	Kill 1	1	Line Down	0	0
06-18 08:17:19	Kill 1	1	Line Running	0	0
06-18 08:19:21	Kill 1	1	Line Down	0	0
06-18 08:20:29	Kill 1	1	Line Running	0	0
06-18 08:28:29	Kill 1	1	Line Empty / No Product	0	0
06-18 08:31:13	Kill 1	1	Product Resuming On Line	0	0

Counters – Reports - Lot Report

Not all installations have the lot report as it only applies to systems with kill line counters.

The Lot report displays the total product count of each lot processed during the production day. An individual lot is the number of birds that were received from a particular growing location. The total count for a lot shown in the report can be compared with the grower's count.

The screenshot shows the 'Lot Bird Count for 08-09-2013' report. The interface includes a sidebar with filters for Shift (All Shifts, Shift 1, Shift 2), Line (Kill Line 1), Time Period (Day, Week, Month, Selected Range), and Date (August 2013). The main table displays the following data:

Lot End Time	Line	Lot Number	Total Count
08-09 06:23:53	Kill 1	1	8694
08-09 10:07:50	Kill 1	2	28129
08-09 12:18:16	Kill 1	3	11407
08-09 23:15:10	Kill 1	4	64347

Counters – Reports - Line Stretch Report

Not all installations will have the line stretch report as it only applies to systems with kill and evis lines where line stretch is enabled.

Line Stretch tracks the number of shackles on a line over time. As the line stretches, shackles are removed to keep it running smoothly. Eventually, enough shackles are removed that a new line is needed to maintain the desired processing speed for birds per minute.

The Line Stretch report displays the number of shackles that were present when the line was new, the number of line shackles currently, the number of line shackles removed since the line was new and the line percent left value. When the line percent left value reaches 0% or less, then the line is considered stretched to the maximum desired amount and needs to be replaced.

The screenshot shows the 'Line Stretch Report for Kill Line on 12-02-2014'. The interface includes a sidebar with filters for Shift (All Shifts, Shift 1, Shift 2), Line (All Lines, Kill Line, Evis Line, Salvage Line), Time Period (Day, Week, Month, Selected Range), and Date (December 2014). The main table displays the following data:

Date	Line	New Line Shackle Count	Current Shackle Count	Shackles Removed	Line Percent Left
12-02-14 00:00	Kill	1907	1907	0	100.0
12-02-14 00:59	Kill	1907	1907	0	100.0
12-02-14 02:00	Kill	1907	1907	0	100.0
12-02-14 02:59	Kill	1907	1907	0	100.0
12-02-14 03:59	Kill	1907	1907	0	100.0
12-02-14 05:00	Kill	1907	1907	0	100.0
12-02-14 05:59	Kill	1907	1907	0	100.0
12-02-14 06:59	Kill	1907	1907	0	100.0
12-02-14 07:59	Kill	1907	1907	0	100.0
12-02-14 09:00	Kill	1907	1907	0	100.0
12-02-14 10:00	Kill	1907	1907	0	100.0
12-02-14 10:59	Kill	1907	1907	0	100.0
12-02-14 12:00	Kill	1907	1907	0	100.0
12-02-14 12:59	Kill	1907	1907	0	100.0
12-02-14 13:59	Kill	1907	1907	0	100.0
12-02-14 14:59	Kill	1907	1907	0	100.0
12-02-14 16:00	Kill	1907	1907	0	100.0
12-02-14 17:00	Kill	1907	1907	0	100.0

Counters - Charts Tab

Select the **Counters** tab, the **Charts subtab**, and the **Counter Data tab** above the report window. The **Charts** subtab will always have the **Counter Data** tab and may include a **Line Stretch Data** tab if line stretch is enabled for one or more lines.

Counters – Charts - Counter Data

The Counter Data chart graphs the line speed and product rate for each minute during the selected shift.

The **Line Speed** is displayed as a **Black** line and the **Product Rate** is displayed as a **Red** line.



Zoom In on the Chart:

1. Click and hold the left mouse button at the top-left corner of the area you want to zoom in on.
2. Drag the mouse down and to the right to create a box around the desired section.
3. Release the mouse button to zoom in.
4. Use the scrollbars to move left, right, up or down.

Exit Zoom Mode:

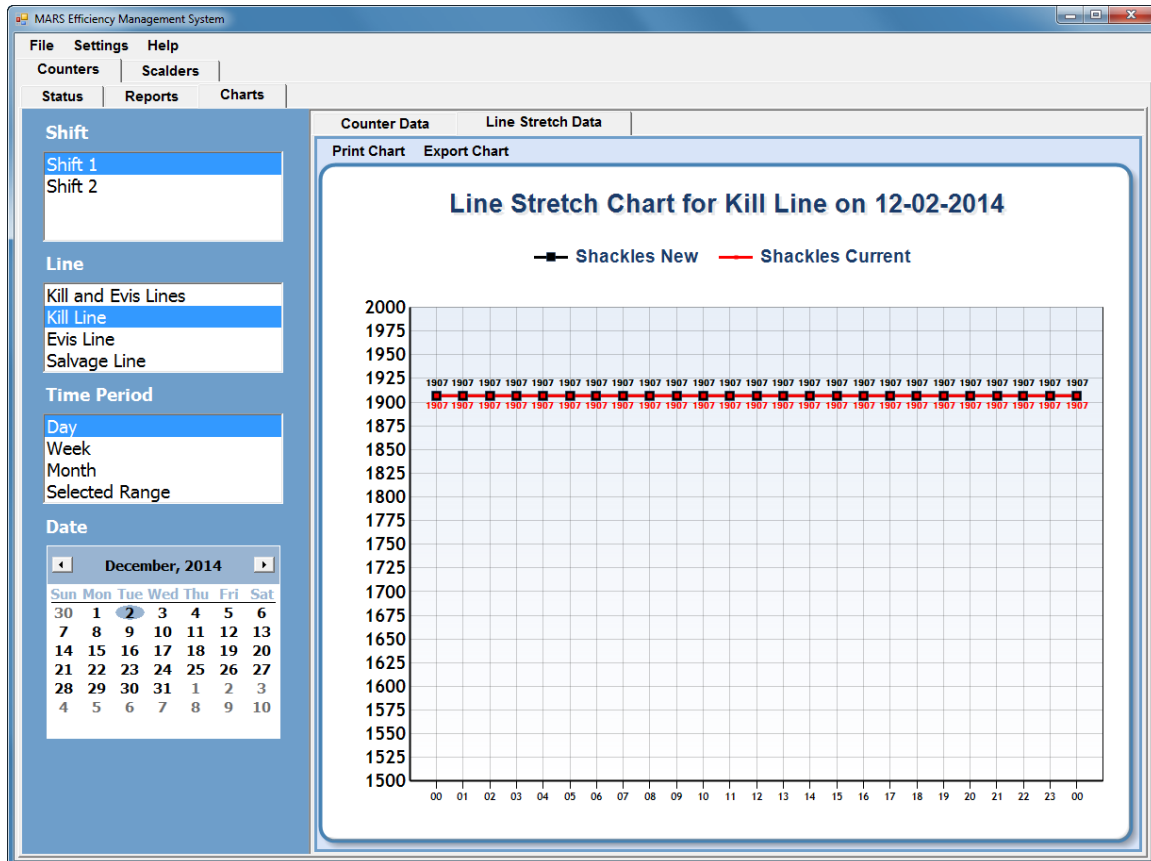
- Press the Escape (Esc) key to return to the full chart view.
- Selecting a different shift, line, or date will also reset the zoom and show a new chart.



Counter – Charts - Line Stretch Data

Select the **Counters** tab, the **Charts** subtab, and the **Line Stretch** tab above the report window. The **Line Stretch Data** tab is only available if line stretch is enabled for one or more lines.

The Line Stretch Data chart graphs the number of shackles when the line was new against the current number of line shackles as they are removed due to line stretch.



Scalders Module

The MARS Efficiency Management System **Scalders Module** communicates with MARS Controllers in the plant that are running the MARS Scalding System. A single MARS Controller can monitor and maintain the temperature of up to four scalding temperature probes and four corresponding steam valves. During the production day, data is continuously uploaded from the controllers to the Windows computer running the MARS Efficiency Management System. The data uploaded is stored in a Microsoft SQL Server database and is made available to the user via a live **Status** tab, a **Reports** tab and a **Charts** tab.

Report Generation and Navigation

The available reports and charts follow the general navigation shown at the beginning of this document. Available reports are listed on the report tab and report filters are on the left blue panel. General windows navigation and printing/exporting options are above the report.

How to enter a range of dates:

To use the **Selected Range** option:

1. Click **Selected Range** under Time Period.
2. Click the start date on the calendar.
3. Hold the **Shift** key and click the end date.

The selected dates will be highlighted, and the report will display data for this range.

Emails and Text Messages

The module can send emails and text messages with the End Of Shift, Lunch, Hourly and Weekly Reports. For detailed setup instructions, refer to the **Email Setup** and **Email Recipient** sections.

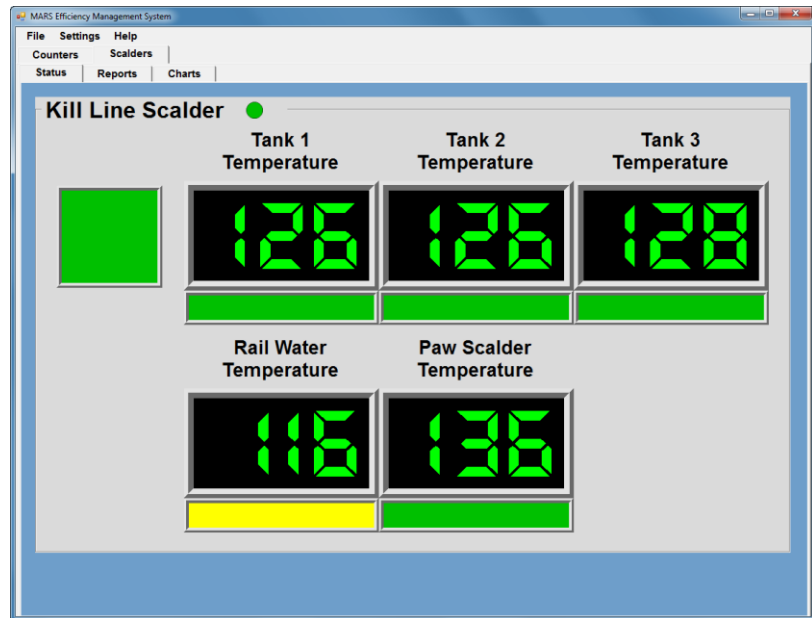
Scalders Status Tab

In this example, the **Scalders** tab is open, and the **Status** subtab is selected. This subtab shows the live status of each scalders in separate blocks.

Each block displays the current temperature and a colored rectangle indicating how close it is to the desired temperature:

- **Green:** Temperature is within acceptable range.
- **Yellow:** Temperature is slightly out of range.
- **Red:** Temperature is unacceptable.

The temperature difference that changes the rectangle to yellow or red is set in the Scalders Line Setup screen.



Each scalders's status is shown in a colored square under its name:

- **Green Square:** Active production is in progress.
- **Blue Square:** The scalders is on break, before a shift, or after a shift.

Status messages may include:

- **"No Comm":** General communication problem with the controller.
- **"Temp Comm":** Issue between the controller and temperature probe module.
- **"Output Comm":** Issue between the controller and output control module.

Next to the scalders name, a round circle shows communication with the MARS controller:

- **Green Circle:** Communication is successful.
- **Red Circle:** There is a communication problem.

Scalders – Report - Shift Report

The **Shift** report displays the average target temperature, actual temperature and output percent of each system for the selected shift, line and time period.

Select report parameters from the blue panel on the left.

Kill Line Scalders Shift 1 Report for Week of 12-01-2014

Date	Line	System	Shift	Target Temperature Average	Actual Temperature Average	Output Percent Average
12-01-2014	Scalders	Tank 1	1	126	127	1.5
12-01-2014	Scalders	Tank 2	1	126	125	21.1
12-01-2014	Scalders	Tank 3	1	128	128	7.9
12-01-2014	Scalders	Rail	1	122	117	0.0
12-01-2014	Scalders	Paw	1	136	133	0.0
12-02-2014	Scalders	Tank 1	1	126	127	4.2
12-02-2014	Scalders	Tank 2	1	126	125	22.8
12-02-2014	Scalders	Tank 3	1	128	128	18.4
12-02-2014	Scalders	Rail	1	122	116	0.0
12-02-2014	Scalders	Paw	1	136	132	0.0
12-03-2014	Scalders	Tank 1	1	126	127	4.8
12-03-2014	Scalders	Tank 2	1	126	126	23.4
12-03-2014	Scalders	Tank 3	1	128	128	19.8
12-03-2014	Scalders	Rail	1	122	115	0.0
12-03-2014	Scalders	Paw	1	136	134	0.0
12-04-2014	Scalders	Tank 1	1	126	127	3.8
12-04-2014	Scalders	Tank 2	1	126	126	21.6
12-04-2014	Scalders	Tank 3	1	128	128	14.7
12-04-2014	Scalders	Rail	1	122	111	0.0

Scalders – Reports - Hourly Report

The Hourly report shows the average target temperature, actual temperature and output percent of each system for each hour during the selected shift.

If there is a scheduled break that begins before the next hour is complete, then the report will display a partial hour ending at the start of the scheduled break.

The next hourly data displayed will then be an hour after the scheduled break ends.

Kill Line Scalders Shift 1 Hourly Report for 12-03-2014

Time	Line	System	Shift	Target Temperature Average	Actual Temperature Average	Output Percent Average
12-03 06:04	Scalders	Tank 1	1	126	127	3.3
12-03 06:04	Scalders	Tank 2	1	126	126	20.4
12-03 06:04	Scalders	Tank 3	1	128	128	14.9
12-03 06:04	Scalders	Rail	1	122	104	0.0
12-03 06:04	Scalders	Paw	1	136	132	0.0
12-03 07:04	Scalders	Tank 1	1	126	127	3.8
12-03 07:04	Scalders	Tank 2	1	126	126	22.1
12-03 07:04	Scalders	Tank 3	1	128	128	17.9
12-03 07:04	Scalders	Rail	1	122	114	0.0
12-03 07:04	Scalders	Paw	1	136	132	0.0
12-03 07:38	Scalders	Tank 1	1	126	127	4.0
12-03 07:38	Scalders	Tank 2	1	126	126	21.8
12-03 07:38	Scalders	Tank 3	1	128	128	18.3
12-03 07:38	Scalders	Rail	1	122	115	0.0
12-03 07:38	Scalders	Paw	1	136	133	0.0
12-03 09:10	Scalders	Tank 1	1	126	127	3.9
12-03 09:10	Scalders	Tank 2	1	126	126	22.5
12-03 09:10	Scalders	Tank 3	1	128	128	18.1
12-03 09:10	Scalders	Rail	1	122	116	0.0
12-03 09:10	Scalders	Paw	1	136	134	0.0
12-03 10:10	Scalders	Tank 1	1	126	127	3.9

MARS Efficiency Management System

File Settings Help

Counters Scalders |

Status Reports Charts

Shift

All Shifts
Shift 1
Shift 2

Line

Kill Line Scalder

System

All Systems
Tank 1
Tank 2
Tank 3
Rail Water
Paw Scalder

Time Period

Day
Week
Month
Selected Range

Date

< December, 2014 >

Sun	Mon	Tue	Wed	Thu	Fri	Sat
30	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	31	1	2	3
4	5	6	7	8	9	10

Kill Line Scalder Shift 2 Detail Report for 12-03-2014

Shift		Hourly	Detail	Events				
Time	Line	System	Shift	BK	Target Temperature	Actual Temperature	Output Percent	
12-03 14:04	Scalder	Tank 1	2		126	127	0.0	
12-03 14:04	Scalder	Tank 2	2		126	126	18.8	
12-03 14:04	Scalder	Tank 3	2		128	128	20.0	
12-03 14:04	Scalder	Rail	2		122	115	0.0	
12-03 14:04	Scalder	Paw	2		136	134	0.0	
12-03 14:05	Scalder	Tank 1	2		126	126	0.0	
12-03 14:05	Scalder	Tank 2	2		126	125	100.0	
12-03 14:05	Scalder	Tank 3	2		128	128	19.4	
12-03 14:05	Scalder	Rail	2		122	115	0.0	
12-03 14:05	Scalder	Paw	2		136	134	0.0	
12-03 14:07	Scalder	Tank 1	2		126	127	0.0	
12-03 14:07	Scalder	Tank 2	2		126	126	0.0	
12-03 14:07	Scalder	Tank 3	2		128	128	18.8	
12-03 14:07	Scalder	Rail	2		122	115	0.0	
12-03 14:07	Scalder	Paw	2		136	134	0.0	
12-03 14:08	Scalder	Tank 1	2		126	127	0.0	
12-03 14:08	Scalder	Tank 2	2		126	126	18.8	
12-03 14:08	Scalder	Tank 3	2		128	128	19.4	
12-03 14:08	Scalder	Rail	2		122	115	0.0	
12-03 14:08	Scalder	Paw	2		136	135	0.0	
12-03 14:09	Scalder	Tank 1	2		126	127	0.0	
12-03 14:09	Scalder	Tank 2	2		126	125	20.0	

Scalders - Reports - Events

- Temperature Probe Burnout
- Temperature Probe OK,
- Temperature Probe Communications Error
- Temperature Probe Communications OK
- Output Communications Error
- Output Communications OK.

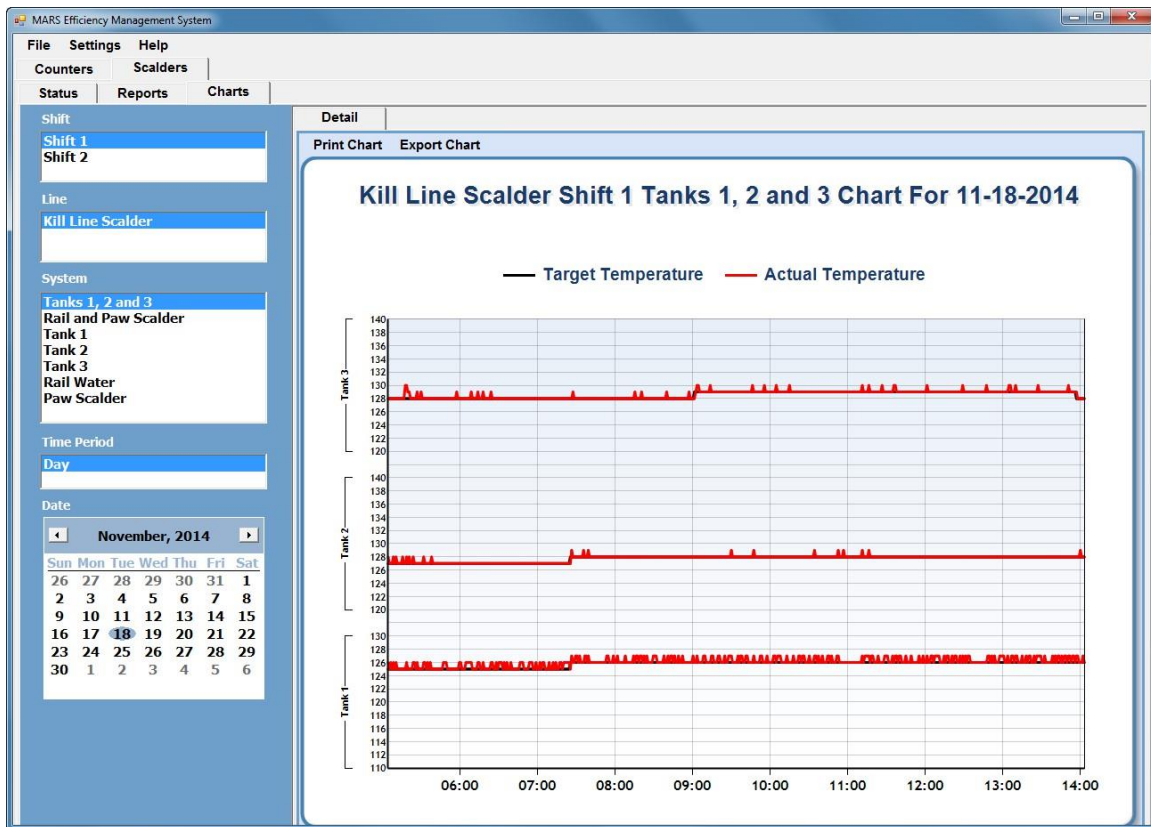
- Controller Reset Success
- Controller Reset Failed
- Controller Reset Detected

[illegible]

Scalders Charts Tab

Select the **Scalders** tab, the **Charts** subtab, and then the **Details** tab above the report window. The Detail chart graphs the target temperatures and actual temperatures for the selected systems for each minute during the selected shift.

The **Target Temperature** is displayed as a **Black** line and the **Actual Temperature** is displayed as a **Red** line.

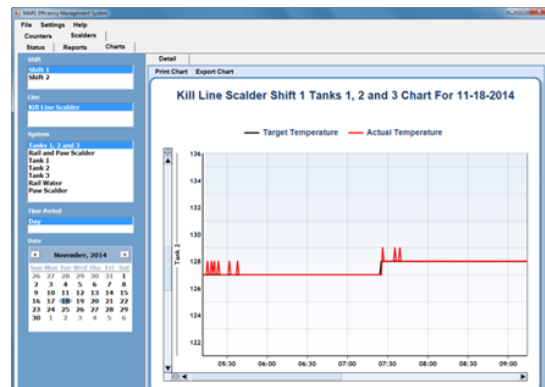


Zoom In on the Chart:

- Click and hold the left mouse button at the top-left corner of the area you want to zoom in on.
- Drag the mouse down and to the right to create a box around the desired section.
- Release the mouse button to zoom in.
- Use the scrollbars to move left, right, up or down.

Exit Zoom Mode:

- Press the Escape (Esc) key to return to the full chart view.
- Selecting a different shift, line, or date will also reset the zoom and show a new chart.



Trimmer Efficiency Module

The MARS Efficiency Management System **Trimmer Efficiency Module** communicates with MARS Controllers in the plant that are running the **MARS Trimmer Efficiency System**. The purpose of the MARS Trimmer Efficiency System is to increase the productivity of your manual trimming operations. This is accomplished by having a separate **Trimmer Hopper** with **Dump Gate** for each trimmer employee on a line. After each piece is trimmed, the trimmer places it in their trimmer hopper. Pieces accumulate in the trimmer hopper until the system dumps the hopper as part of a cycle that gets repeated continuously during the shift. The Trimmer Efficiency System can measure the efficiency of up to twenty trimmers with a single MARS Controller. During the production day, data is continuously uploaded from the controllers to the Windows computer running the MARS Efficiency Management System. The data uploaded is stored in a Microsoft SQL Server database and is made available to the user via a live **Status** tab, a **Reports** tab and a **Charts** tab.

Report Generation and Navigation

The available reports and charts follow the general navigation shown at the beginning of this document. Available reports are listed on the report tab and report filters are on the left blue panel. General windows navigation and printing/exporting options are above the report.

How to enter a range of dates:

To use the **Selected Range** option:

1. Click **Selected Range** under Time Period.
2. Click the start date on the calendar.
3. Hold the **Shift** key and click the end date.

The selected dates will be highlighted, and the report will display data for this range.

Emails and Text Messages

The module can send emails and text messages with the End Of Shift, Lunch, Hourly and Weekly Reports. For detailed setup instructions, refer to the **Email Setup** and **Email Recipient** sections.

TRIMMER EFFICIENCY STATUS TAB

Trimmer Efficiency Status Tab

In the screen below, **Trimmer Efficiency** is the primary tab selected, and **Status** is the subtab selected. The **Status** subtab displays the live Status of each Trimmer Line in a separate block. The data displayed in the block for each trimmer line includes the **Incoming Total Pounds**, the **Outgoing Total Pounds** and the **Percent Yield**. The percent yield is calculated by dividing the outgoing total pounds by the incoming total pounds and multiplying the result by 100. The block for each trimmer line also displays the **Total Pounds** trimmed and latest **Pounds Per Minute** value for each trimmer.

There is a cumulative totals block for all trimmer lines in the lower right.

This block totals the incoming weights, outgoing weights, trimmer total pounds and trimmer pounds per minute for all of the trimmer lines.



An overall percent yield is also calculated and displayed.

The general trimmer line status is displayed in a colored square under the trimmer line name. The square is **Green** if there is an active production period in progress and is **Blue** if the trimmer line is currently on break, before the start of a shift or after the end of a shift. The communications status with the MARS controller is also displayed for each trimmer line as a round circle to the right of the trimmer line name. If the circle is **Green** communication is successful and if the circle is **Red** there is currently a communications problem between the MARS Efficiency Management System software and the controller.

Trimmer Efficiency – Reports - Shift Report

The Shift report shows the average pounds per minute trimmed and total pounds trimmed for the selected shift, line, trimmer and time period.

The screenshot shows the MARS Efficiency Management System interface. The left sidebar contains filters for Shift (All Shifts, Shift 1, Shift 2), Line (All Lines, Meyn 1 Trimmer, Meyn 2 Trimmer, Meyn 3 Trimmer), Trimmer (All Trimmers, Trimmer 1, Trimmer 2, Trimmer 3, Trimmer 4, Trimmer 5, Trimmer 6), and Date Period (Day, Week). The main area displays the 'Meyn 3 Trimmer Shift 1 Report for 04-08-2013' with a table of data.

Date	Line	Trimmer	Shift	Lbs/Minute	Total Lbs
04-08-2013	Meyn 3	1	1	6.9	3,409
04-08-2013	Meyn 3	2	1	3.6	1,757
04-08-2013	Meyn 3	3	1	4.1	1,994
04-08-2013	Meyn 3	4	1	4.0	1,956
04-08-2013	Meyn 3	5	1	7.3	3,609
04-08-2013	Meyn 3	6	1	4.6	2,243
04-08-2013	Meyn 3	7	1	4.4	2,144
04-08-2013	Meyn 3	8	1	3.5	1,709
04-08-2013	Meyn 3	All	1	4.8	18,821

Trimmer Efficiency – Reports - Hourly Report

The Hourly Report shows:

- **Hourly Average Pounds/Minute:** Average for each hour.
- **Hourly Pounds Trimmed:** Pounds trimmed in each specific hour.
- **Total Average Pounds/Minute:** Overall average for the shift.
- **Total Pounds Trimmed:** Cumulative total for the shift.

Hourly values are specific to each hour, while total values cover the entire shift.

If a scheduled break starts before an hour is complete, the report will show a partial hour ending at the break's start. The next hour's data will appear an hour after the break ends.

The screenshot shows the MARS Efficiency Management System interface. The left sidebar contains filters for Shift (All Shifts, Shift 1, Shift 2), Line (All Lines, Meyn 1 Trimmer, Meyn 2 Trimmer, Meyn 3 Trimmer), Trimmer (All Trimmers, Trimmer 1, Trimmer 2, Trimmer 3, Trimmer 4, Trimmer 5, Trimmer 6), and Date Period (Day, Week). The main area displays the 'Meyn 3 Trimmer Hourly Report for 04-17-2013' with a table of data.

Time	Line	Trimmer	Shift	Hourly Lbs Per Minute	Hourly Lbs	Total Lbs Per Minute	Total Lbs
04-16 23:50	Meyn 3	5	1	4.6	274	4.6	274
04-17 00:50	Meyn 3	5	1	4.8	287	4.7	562
04-17 01:40	Meyn 3	5	1	6.1	304	5.1	866
04-17 03:13	Meyn 3	5	1	6.3	376	5.4	1,242
04-17 04:13	Meyn 3	5	1	6.3	377	5.6	1,619
04-17 04:20	Meyn 3	5	1	3.5	25	5.5	1,644
04-17 05:53	Meyn 3	5	1	6.1	365	5.6	2,009
04-17 06:53	Meyn 3	5	1	5.3	316	5.6	2,325
04-17 07:53	Meyn 3	5	1	4.5	268	5.4	2,593
04-17 08:08	Meyn 3	5	1	5.7	87	5.4	2,680

Trimmer Efficiency – Reports - Details Report

The Detail report displays the pounds per minute trimmed and total pounds trimmed for each minute during the selected shift. During scheduled breaks an X will be displayed in the column named BK.

Meyn 3 Trimmer Detail Report for 04-17-2013

Time	Line	Trimmer	Shift	BK	Lbs/Minute	Total Lbs
04-17 00:26:58	Meyn 3	5	1		4.6	454.5
04-17 00:28:03	Meyn 3	5	1		4.7	462.3
04-17 00:29:08	Meyn 3	5	1		4.5	464.8
04-17 00:30:13	Meyn 3	5	1		4.4	468.9
04-17 00:31:15	Meyn 3	5	1		4.7	481.4
04-17 00:32:20	Meyn 3	5	1		4.7	488.1
04-17 00:33:25	Meyn 3	5	1		5.8	495.3
04-17 00:34:30	Meyn 3	5	1		5.6	497.6
04-17 00:35:35	Meyn 3	5	1		5.5	502.4
04-17 00:36:41	Meyn 3	5	1		5.3	508.1
04-17 00:37:46	Meyn 3	5	1		5.3	509.5
04-17 00:38:51	Meyn 3	5	1		5.2	517.8
04-17 00:39:56	Meyn 3	5	1		5.3	519.9
04-17 00:41:01	Meyn 3	5	1		4.9	524.8
04-17 00:42:07	Meyn 3	5	1		4.3	527.7
04-17 00:43:12	Meyn 3	5	1		4.0	533.3
04-17 00:44:17	Meyn 3	5	1		3.9	536.1
04-17 00:45:23	Meyn 3	5	1		3.8	540.0
04-17 00:46:28	Meyn 3	5	1		4.3	548.3
04-17 00:47:34	Meyn 3	5	1		4.1	550.7
04-17 00:48:39	Meyn 3	5	1		4.0	558.1
04-17 00:49:44	Meyn 3	5	1		4.4	561.6
04-17 00:50:45	Meyn 3	5	1		5.3	591.5
04-17 00:51:50	Meyn 3	5	1		6.7	607.6

Trimmer Efficiency - Reports - Events

The Controller Reset events include:

Controller Reset Success
Controller Reset Failed
Controller Reset Detected.

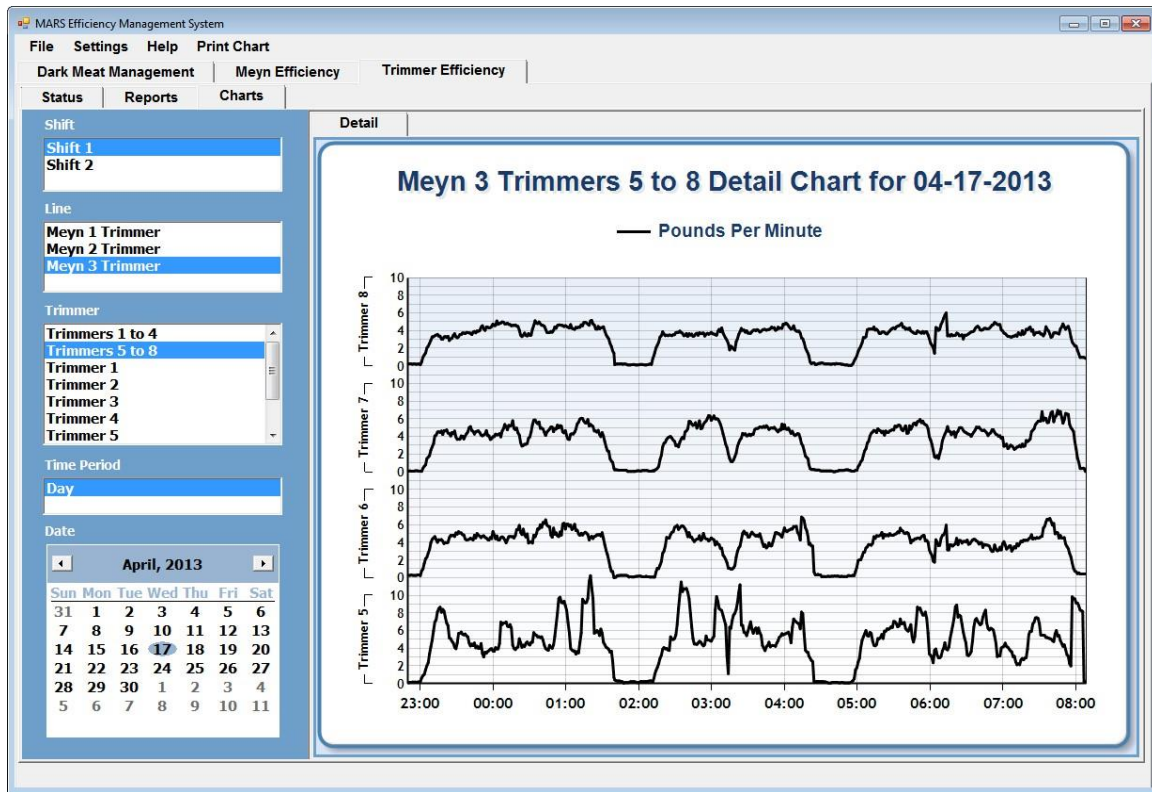
Meyn 3 Trimmer Events For 04-17-2013

Time	Line	Shift	Description	Data1	Data2
04-16 22:44:06	Meyn 3	1	Controller Reset Success	3	0

The checkboxes above the report control which events to displayed. before the beginning of each shift.

Trimmer Efficiency - Charts - Details

The **Pounds Per Minute** trimmed is displayed as a **Black** line.



Zoom In on the Chart:

9. Click and hold the left mouse button at the top-left corner of the area you want to zoom in on.
10. Drag the mouse down and to the right to create a box around the desired section.
11. Release the mouse button to zoom in.
12. Use the scrollbars to move left, right, up or down.

Exit Zoom Mode:

- Press the Escape (Esc) key to return to the full chart view.
- Selecting a different shift, line, or date will also reset the zoom and show a new chart.

